

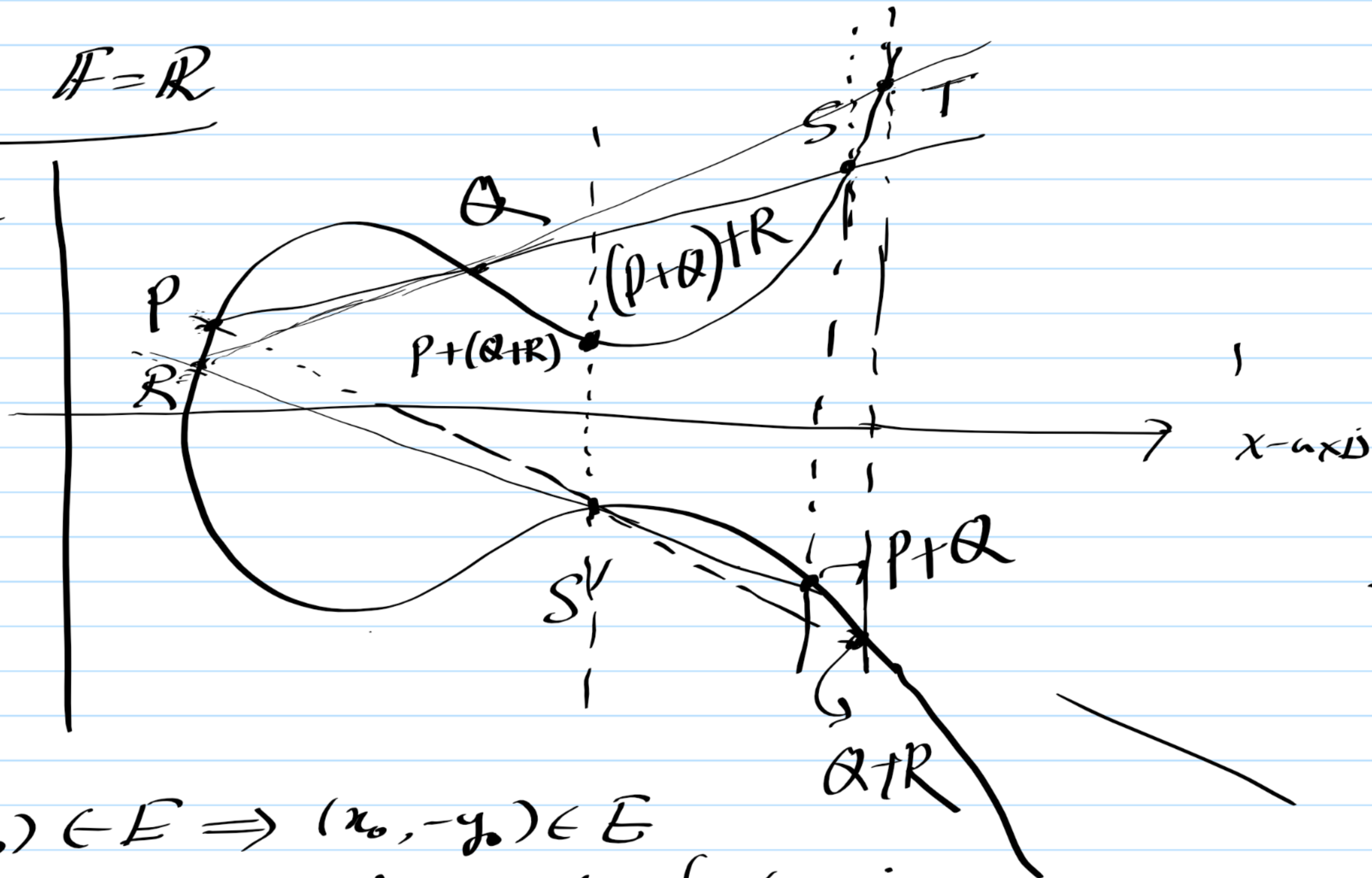
Elliptic curves

Consider $y^2 = x^3 + ax + b$

$$E = \{ (x, y) \in \mathbb{F}^2; y^2 = x^3 + ax + b \} \cup \{\infty\}$$

For example $\mathbb{F} = \mathbb{R}$

Place where the curve in $\mathbb{R}^2$ has the form as in the diagram.



$$\text{If } (x_0, y_0) \in E \Rightarrow (x_0, -y_0) \in E$$

So E is symmetric about x-axis.

Let $P, Q \in E \setminus \{\infty\}$

Join P & Q , the line intersects E at a third point S

Then join S & ∞ & the line intersects E at one more pt call it $P+Q$

Let $R \in E$, find $(P+Q)+R$ &
 $P+(Q+R)$

& compare & see that they match

This needs Max Noether theorem of Algebraic geometry.

Observe that $P + \infty = P$ $\forall P$

$$P + (-P)$$

$$P + S = \infty$$

as when we join $P = (x_0, y_0)$

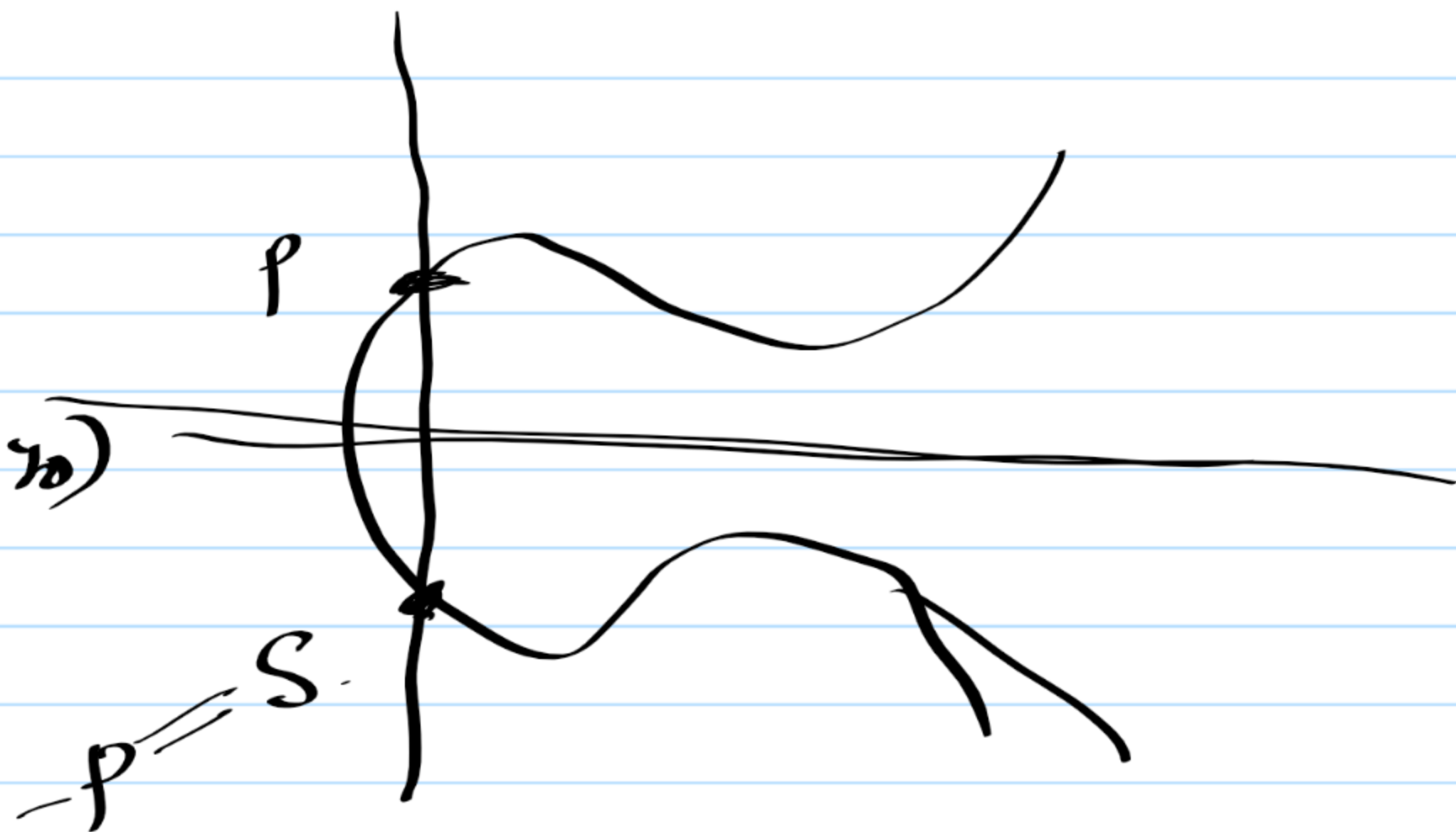
$$\& -P = (x_0, -y_0)$$

the third pt is ∞

& join ∞ with ∞

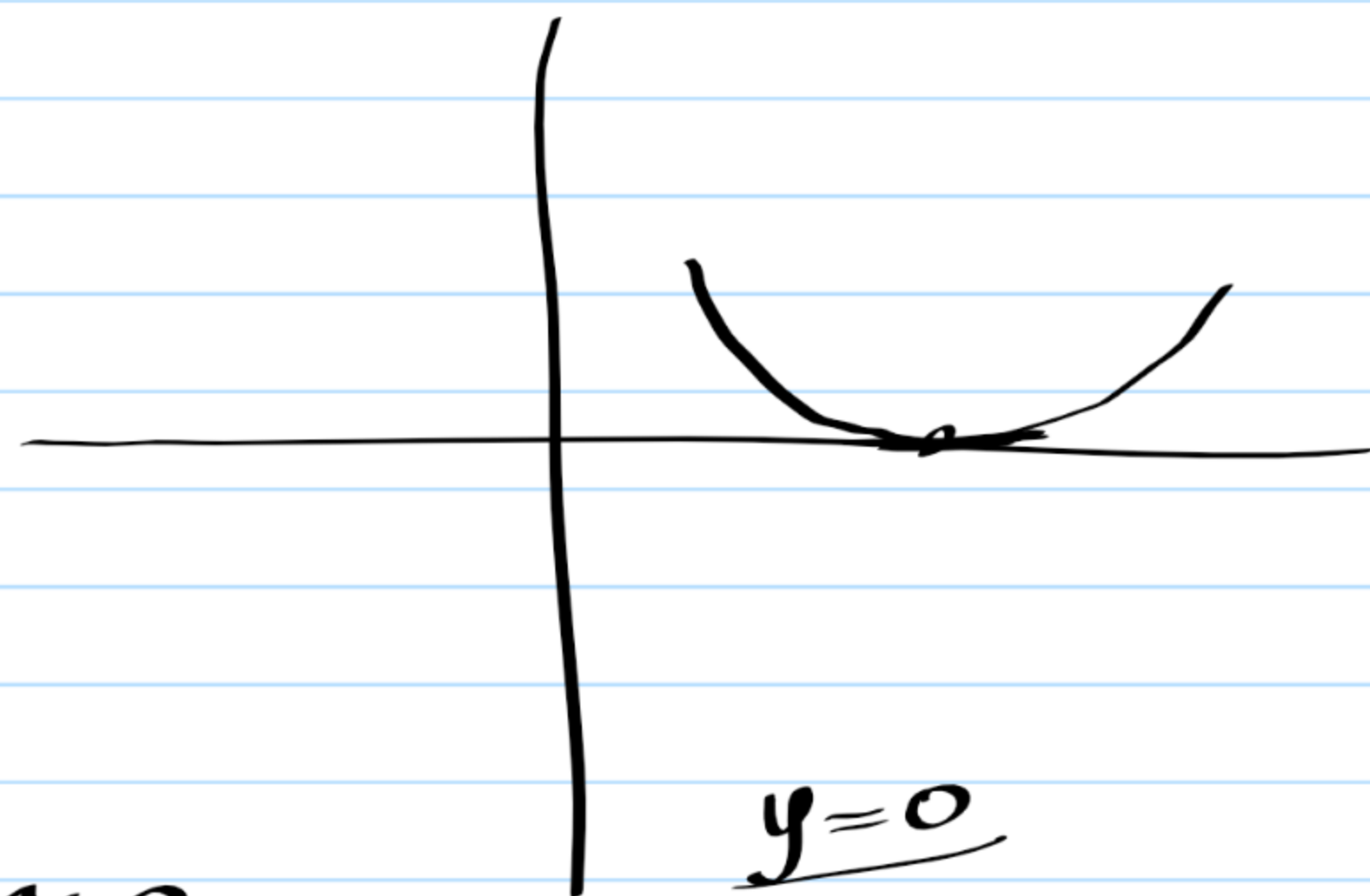
i.e. the tangent at ∞ , then the third pt is ∞

$$\boxed{P + (-P) = \infty}$$



$$\begin{aligned} & \text{in } \mathbb{P}R^2 \\ & \infty = [(0, 1, 0)] \\ & 2y^2 = x^3 + axz^2 + bz^3 \end{aligned}$$

$$\begin{aligned} & \text{put } x=0, \quad 2y^2 = bz^3 \\ & \left. \begin{aligned} x=0, \quad 2(y^2 - bz^3) &= 0 \\ 2(1 - bz^2) &= 0 \\ z=0, \text{ or } z^2 &= \pm \frac{1}{b} \end{aligned} \right\} \end{aligned}$$



Singular & nonsingular curves.

A point $P = (x_0, y_0)$ on a curve $f(x, y) = 0$ is called a singular pt if

$$\frac{\partial f}{\partial x}(P) = 0 \quad \&$$

$$\frac{\partial f}{\partial y}(P) = 0$$

Recall

$$\frac{dy}{dx}(P) = - \frac{\frac{\partial f}{\partial x}(P)}{\frac{\partial f}{\partial y}(P)} \quad \checkmark$$

$$\left[\begin{array}{l} f(x,y)=0 \Rightarrow f_x + f_y \cdot \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{f_x}{f_y} \\ x=x(t) \\ y=y(t) \end{array} \right]$$

$$\underline{f = x^2 + y^2 - 1}$$

$f_x = 0 = f_y \Rightarrow (x,y) = (0,0)$ which is not on the curve

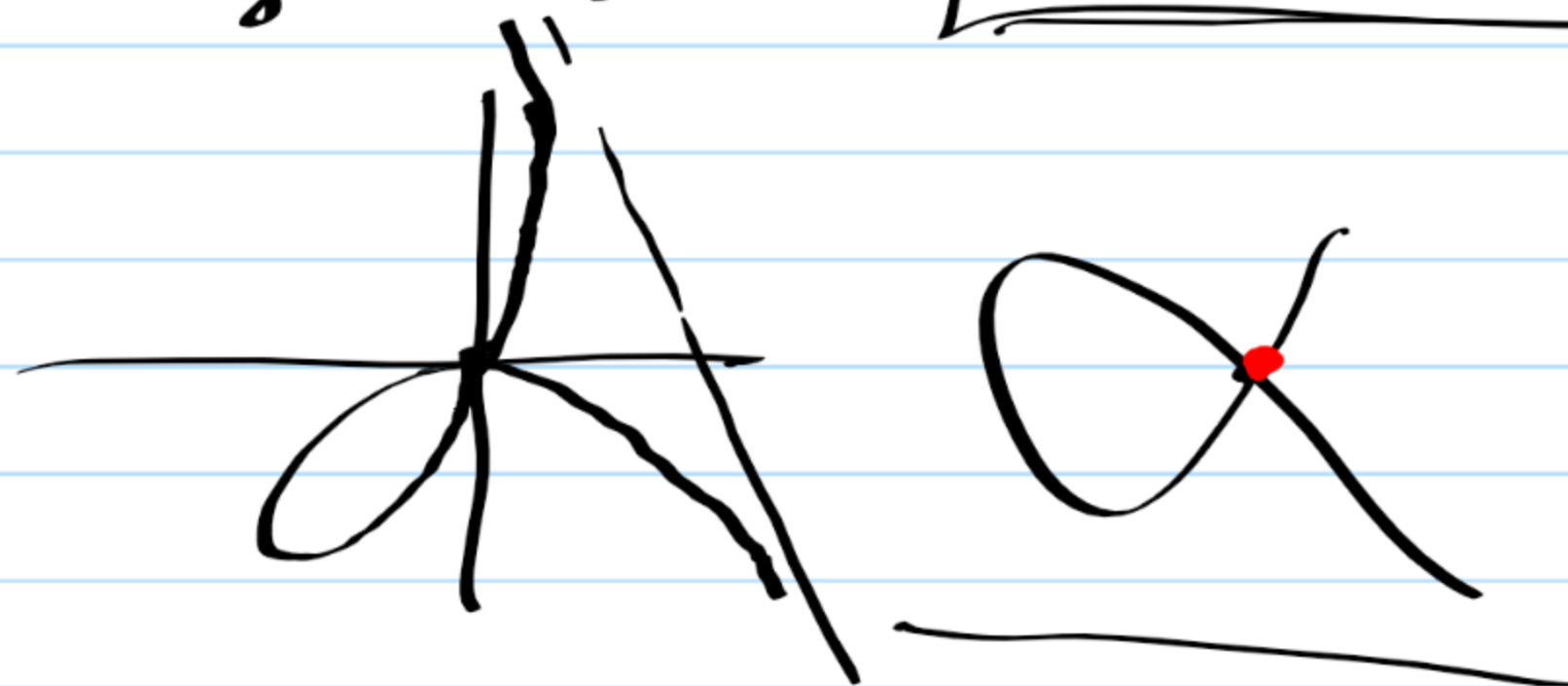
so $x^2 + y^2 = 1$ is a non singular curve.

$$\bullet f = (x-1)^2 - (y-1)(y-2)^2$$

Solve $\underline{f_x = 0 = f_y}$

Solns $(1,2)$ & $(1, \frac{4}{3})$ are possibilities.

so $(1,2)$ is the only singular pt. $\boxed{(1, \frac{4}{3}) \notin C}$



Ex: $\underline{x^3 + y^3 + 3xy = 0}$

$O = (0,0)$ lies on the curve.

& the tangents at $(0,0)$ are given by the least order terms, i.e. $3xy = 0$ i.e. $x=0$ & $y=0$.

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \text{sup } P = (a, 0)$$

$\nabla f((x-a), y)$, the least degree term gives you the tangent at $(a,0)$

$$\checkmark \frac{x_1 x}{a^2} + \frac{y_1 y}{b^2} = 0$$

$$\frac{x}{a} + 0 = 0 \quad \therefore \underline{x=0}$$

Double pt, Triple pt, multiple pt.

↓
two branches ↓
3 branches so on.